

Systems Analysis and Design

Lec 1: The Systems Development Environment

*What is Systems Analysis and Design?

The process of developing and maintaining an information system.

The systems analyst to succeed has to:

1. Understand the organisation objectives, structure and processes.
2. Have knowledge how to use IT for the advantage of the organisation. In addition to following a structured approach for development (SDLC)

*Systems Analysis & Design Core Concepts

The main goal is developing application software to support the organisation function such as Payroll or market analysis.

Application Software: is software designed to process data in an organisation to help users. Examples are spreadsheets, word processor and DBMS.

Core Concepts cont.

*Software Engineering Concepts.

Methodologies: Are a sequence of step-by step approaches that help in developing the information systems. It can include several techniques such as interviews.

Techniques: Are processes that the system analyst use for developing the system.

Tools: Are software programs such as CASE tools used to make using the techniques easier.

*Systems and its Components

Definition: Is a set of inter-related business procedures or components within one business unit for some purpose.

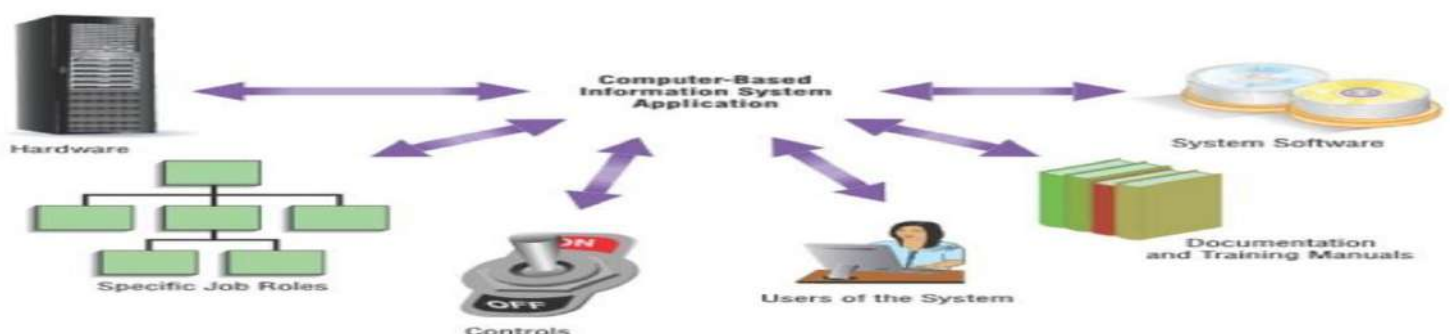
Characteristics of a system are:

- | | | | |
|---|--|--------------|-------------|
| 1. Components. | 2. Interrelated Components. | 3. Boundary. | 4. Purpose. |
| 5. Environment. | 6. Interfaces. (Point of contact with environment) | | |
| 7. Constraints. (Limit on what system can do) | 8. Inputs. | 9. Outputs. | |

*Important Systems Concepts

- **Decomposition:** Is the process of breaking the system into smaller components.
- **Modularity:** dividing the system into chunks or modules for easier management.
- **Coupling:** To what extent the subsystems are dependent on each other.
- **Cohesion:** To what extent a subsystem performs a single function.

Components of CBIS



*Components of CBIS:

4- Users of the System

1- Hardware

5- Documentation and Training Manuals

2- Specific Job Roles

3- Controls

6- System Software

*The System Analyst Role

- The main role of a system analyst is studying the **problems** and **needs** of the organisation, to determine how to use the people, methods and IT for the benefit of the organisation.
- The system Analyst helps system users and managers to specify their needs for improving existing systems or creating new systems.

*Skills required for the system Analyst.

- **Analytical**: Helps understand the organisation and its functions.
- **Technical**: Helps understand the capabilities and limitations of IT. i.e. programming and hardware and OS.
- **Managerial**: Helps in project management, risk management and managing resources.
- **Interpersonal**: Helps in dealing with users and programmers.

*Developing IS and the SDLC

System Development Methodology: Is a standard process followed by an organisation to analyse, design, implement and maintain IS.

System Development Life Cycle: The series of steps used to mark the phases of development for an information system.

*Systems Development Life Cycle (SDLC): Four Phases:

*Phase 1: Systems Planning and Selection.

The first phase in the SDLC, in which an organisation's total information system needs are analysed and arranged, and in which a potential information systems project is identified and an argument for continuing or not continuing with the project is presented.

The organisation IS needs results from:

- Solve an existing problem.
- Perform extra tasks.
- Utilise IT for improvement of existing procedures. Feasibility study: determine the economic and organisational impact of the system.

*Phase 2: Systems Analysis

The second phase of the SDLC, in which the current system is studied and proposed replacement system is presented.

The output of the analysis phase is a description of the alternative solution proposed by the analysis team.

*Phase 3: Systems Design

The third phase of the SDLC in which, the proposed system from the analysis phase is first described independently from any technology (HW or SW) called (Logical Design) and then transformed to technology-specific details (Physical Design). Physical design is then used for system construction and implementation.

*Phase 4: Systems Implementation and Maintenance

The final phase of the SDLC, in which the final system is coded, tested and installed in the organisation, and in which the information system is systematically and improved.

It is divided into two stages:

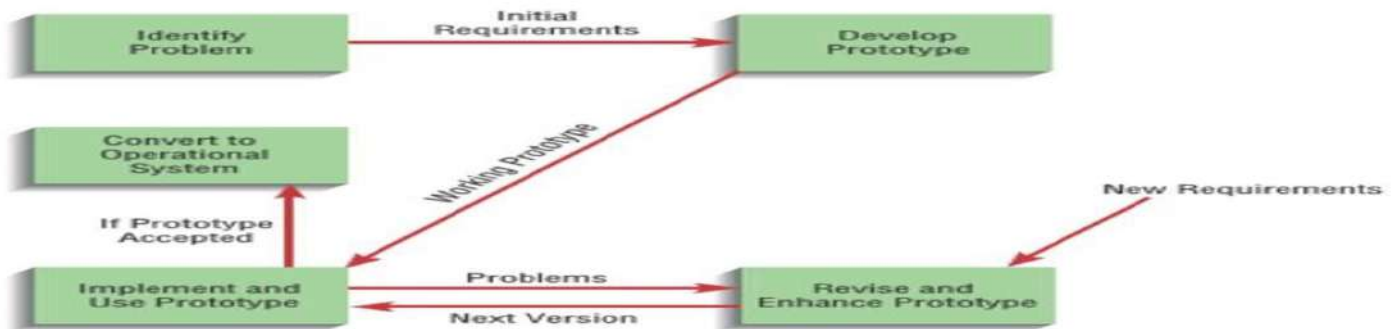
- System Implementation.
- System Operation.

*Alternative Approaches to Development

Prototyping: is building a scaled-down version of the desired information system. **Advantages of prototyping**:

- Involve the users in the analysis and design.
- Capture system requirements in a concrete rather than a verbal way.

Prototyping cont.



*Computer-aided Software Engineering (CASE tools)

Software tools that provide automated support for some portions of systems development.

CASE tools include:

- Diagramming tools.
- Documentation generators
- Computer display and report generators,
- Code generators.

*Joint Application Design (JAD)

-A structured process in which users, managers and analysts work together for several days in a series of intensive meetings to specify or review systems requirements.

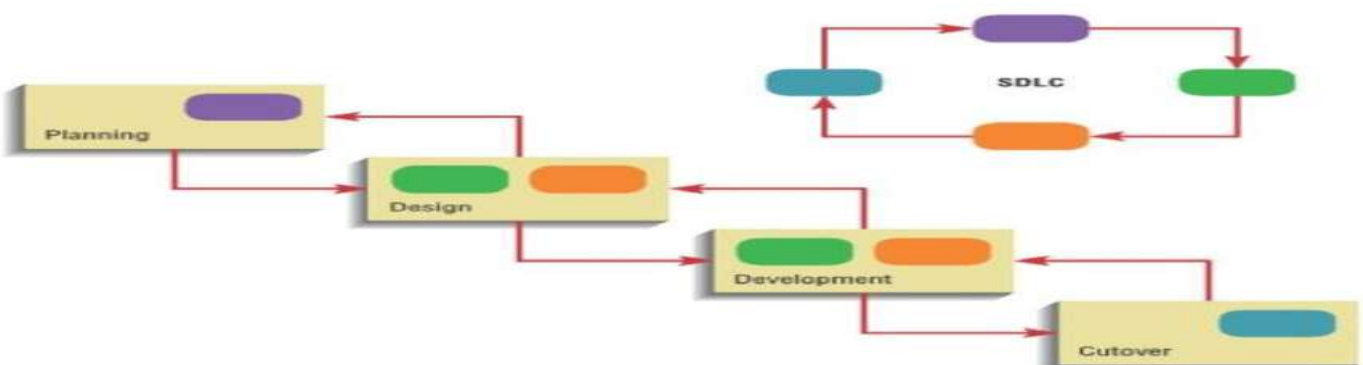
-It is popular in certain applications such as insurance.

*Rapid Application Development (RAD)

A system development methodology created to radically decrease the time needed to design and implement IS.

Prototyping, CASE tools and JAD are used to support RAD.

RAD Vs SDLC



*Participatory

Design A system development methodology originated in Northern Europe, in which users and the improvement of their work lives are the central focus.

*Agile Methodologies

Is a family of development methodologies characterised by short iterative cycles and extensive testing; active involvement of users for establishing, prioritising and verifying requirements and a focus on small teams of talented, experienced programmers.